



**Verified Carbon
Standard**

DEVELOPING CARBON FINANCE PROJECT FOR FARMERS ADOPTING REGENERATIVE AGRICULTURE PRACTICES IN THE INDO-GANGETIC PLAINS



Document Prepared by: Varaha ClimateAg Private Limited

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CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	6
1.3	Project Eligibility	6
1.4	Project Design	6
1.5	Project Proponent	8
1.6	Other Entities Involved in the Project	8
1.7	Ownership.....	8
1.8	Project Start Date	8
1.9	Project Crediting Period	9
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	9
1.11	Description of the Project Activity	10
1.12	Project Location	15
1.13	Conditions Prior to Project Initiation	15
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks	19
1.15	Participation under Other GHG Programs	20
1.16	Other Forms of Credit.....	20
1.17	Sustainable Development Contributions	21
1.18	Additional Information Relevant to the Project	22
2	SAFEGUARDS	23
2.1	No Net Harm	23
2.2	Local Stakeholder Consultation	23
2.3	Environmental Impact	27
2.4	Public Comments	27
2.5	AFOLU-Specific Safeguards	28
3	APPLICATION OF METHODOLOGY.....	28
3.1	Title and Reference of Methodology	28
3.2	Applicability of Methodology	28
3.3	Project Boundary	30
3.4	Baseline Scenario	33
3.5	Additionality	35

3.6	Methodology Deviations	37
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	37
4.1	Baseline Emissions	37
4.2	Project Emissions	37
4.3	Leakage.....	37
4.4	Net GHG Emission Reductions and Removals	38
5	MONITORING	38
5.1	Data and Parameters Available at Validation	38
5.2	Data and Parameters Monitored.....	38
5.3	Monitoring Plan	38
APPENDIX		39

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project intends to generate additional income to the small holder farmers of Indo- Gangetic plain region for adopting improved agricultural practices such as zero till farming, cover cropping, crop diversification, farm residue incorporation, optimizing fertilizer application and efficient water management which provide a potential to not just prevent emissions into the atmosphere but also create carbon sinks through sequestration of carbon back into the soil. Globally, agriculture and allied, account for almost 22% of the total greenhouse gas (GHG) emissions, and hence is considered a major problem contributing to climate change¹.

In 2018, global emissions due to agriculture (within the farm gate and including related land use/land use change) were 9.3 billion tonnes of CO₂ equivalent (CO₂eq). Out of this, India is the third-largest emitter of greenhouse gases and accounted for about 650 Mt CO₂ eq. of the total global emissions (FAO 2018). These high emissions are largely attributed to conventional practices such as flooded, transplanting method of cultivation in rice, multiple tillage operations, residue burning, flooding, monocropping, excessive usage of fertilizers and water, and absence of cover crops for Carbon (C) and Nitrogen (N) management in soils. The average Soil Organic Carbon (SOC) content in these soils is around 0.2–0.4% (MoA & FW 2019) as against the ideal (Bhattacharya, 2007).

In November 2021, India announced its new climate targets at COP26, Glasgow under its 'Panchamrit (or a 5 elixir) strategy'. Of these five strategies, three focused explicitly on reducing Carbon emissions.² In order to curb the current extent of agricultural GHG emissions (national level ~14%) and achieve these ambitious goals, business-as-usual approach would not suffice, and India must intensively strategize towards development of low-emission agriculture pathways.

Despite its appeal, regenerative agriculture will take a long time to scale across Indian geographies especially because there is a lack of incentive to invest in regenerative practices with a long-term return on investment. The newly emerging carbon market for agriculture is envisioned to stimulate low-emission agricultural technologies. However, the transition to negative carbon agriculture requires a strong value proposition.

In this context, the project implements faster adoption of improved agricultural practices by Indian farmers that can be the only solution to mitigate the GHG emissions, reverse land degradation and enable carbon sequestration, along with improving overall soil health and crop productivity. The benefits of regenerative agriculture to soil fertility are proven through extensive

¹ <https://ourworldindata.org/ghg-emissions-by-sector>

² <https://pib.gov.in/PressReleasePage.aspx?PRID=1768712>

research in India as well as globally³. For example, peer-reviewed research (Jat et. al., 2020)⁴ suggested that direct seeding of rice has a sequestration potential of 500 Kg Co₂ eq/ Ha and zero-till sequesters ~900 Kg Co₂ eq/ Ha in maize and ~1600 Kg Co₂/Ha in wheat, and cover cropping sequesters ~1000 Kg Co₂ eq / Ha.

On the other hand, crop residue burning releases CO₂, CO and NO_x emissions with a Global Warming Potential (GWP) for wheat (1622), Paddy (745), maize (997) and Sugarcane (559) and CH₄ with a GWP for wheat (137), Paddy (186), maize (64) and Sugarcane (36)⁵. Prevention of residue burning will stop these emissions. The reduction potential of laser leveling ranges between 1000 – 5000 Kg CO₂/Ha. In addition to environmental benefits, adoption of regenerative practices also improves incomes for farmers along with increased productivity. For example, in the case of the Rice-Wheat-Maize cropping system, in smallholder farming countries, a net income increase of 12.5% -16% was reported, depending on the change in practices of the farmers. This increase is due to a) increase in productivity over a period, b) income earned from carbon credit sales, and c) a drop in the cost of production (Jat et. Al.,2020)⁶. The current project is being implemented in the Indo Gangetic plains of the Indian subcontinent focused towards promoting regenerative practices among the rice, wheat, maize, cotton, mustard, pulses, and sugarcane farms based in Haryana, Punjab, Uttar Pradesh, Odisha, Uttarakhand, West Bengal, and Bihar. These crops together cover >80% area of cropping in these states with an overall cropped area of 63,924 thousand hectares⁷.

The project will be a grouped project, with new project activities being added over the lifespan of the project. Starting with rice-wheat cropping system, the project will be further expanded to maize-wheat, sugarcane, and cotton-wheat/cotton cropping systems. Pulses and Mustard are cover crops widely interspersed in these cropping systems. The project considers 2019 as the baseline year in which the interventions started. Starting with 12 thousand hectares (~28000 Tonnes CO₂ Eq. per annum) in 2019, the area will be stabilized at 2 million hectares (~7.0 million Tonnes CO₂ Eq. per annum) in 2027 and maintained until 2038.

A small percentage of farmers have started implementing one or more of the regenerative agricultural practices and the number of farmers participating is projected to grow over the course of the project crediting period. VarahaAg Carbon Farming Program is being implemented for scaling up different regenerative agricultural practices which offer significant potential to mitigate climate change and deliver other benefits, promoting wide-scale implementation of agricultural emissions mitigation. The detailed steps of the program are provided in section 1.11.

³ <https://www.ceew.in/publications/sustainable-agriculture-india#:~:text=Natural%20farming%20is%20the%20fastest,after%20decades%20of%20sustained%20promotion>

⁴ Jat, H. S., Kumar, V., Datta, A., Choudhary, M., Kakraliya, S. K., Poonia, T., ... & Sharma, P. C. (2020). Designing profitable, resource use efficient and environmentally sound cereal-based systems for the Western Indo-Gangetic plains. *Scientific reports*, 10(1), 1-16.

⁵ Sahu, S. K., Mangaraj, P., Beig, G., Samal, A., Pradhan, C., Dash, S., & Tyagi, B. (2021). Quantifying the high-resolution seasonal emission of air pollutants from crop residue burning in India. *Environmental Pollution*, 286, 117165.

⁶ Jat, H. S., Kumar, V., Datta, A., Choudhary, M., Kakraliya, S. K., Poonia, T., ... & Sharma, P. C. (2020). Designing profitable, resource use efficient and environmentally sound cereal-based systems for the Western Indo-Gangetic plains. *Scientific reports*, 10(1), 1-16.

⁷ https://www.nfsm.gov.in/ReadyReckoner/CU1/CroppingIntensity_Jan2021.pdf.

The incentives generated from carbon finance projects will create an additional source of income for farmers by ensuring motivation and consistency in adoption of improved practices that will further enhance soil health and climate resilience and thus yield better crop benefits and enriched livelihoods.

1.2 Sectoral Scope and Project Type

The sectoral scope of the project is VCS scope 14 “Agriculture, Forestry and Other Land Use” (AFOLU) and under the category Agriculture Land Management (ALM). The project constitutes activities that reduce as well as remove carbon dioxide from the atmosphere. It falls under the category of grouped project and is using the VCS methodology: “VM0042, Ver 1.0, Methodology for Improved Agricultural Land Management “.

1.3 Project Eligibility

The project will implement sustainable improved agricultural practices. The project is eligible under the scope of the VCS Program. The grouped project is eligible as it fulfills the rules and requirements of the VCS Program criteria for GHG projects and programs.

The project is being submitted under the VM0042 methodology for Improved Agricultural Land Management Version 1.0.

The following conditions describe the compliance with the applicability conditions for the use of the VCS methodology VM0042 methodology for Improved Agricultural Land Management Version 1.0.-

- a) Project activities are implemented on land that is either cropland or grassland at the project start date and remains cropland or grassland throughout the project crediting period (i.e., land use change is not eligible, including conversion from cropland to grassland and grassland to cropland).
- b) The project area has not been cleared of native ecosystems within the 10-year period prior to the project start date.
- c) The project activity is not expected to result in a sustained reduction greater than 5% in productivity, as demonstrated by peer-reviewed and/or published studies on the activity in the region or a comparable region.
- d) The project activity does not involve the application of biochar.
- e) The project activity does not occur on a wetland, however on crops subject to artificial flooding where it is proven that crop cultivation does not impact the hydrology of any nearby wetlands.

All the activities being promoted in this project were shown in peer-reviewed scientific studies to successfully sequester soil organic carbon and mitigate gaseous emissions resulting from adoption of improved agricultural management practices included in the project description.

1.4 Project Design

The project is a grouped project. It includes multiple locations and project activity instances. The geographic boundaries have been well defined, including where new project activity instances

may be added. These establish the criteria for determining the eligibility of future instances. New project activity instances will continue to be added during the verification event.

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

The planned project activities fall under the VCS VM0042 Methodology. This methodology quantifies the GHG emission reductions and SOC resulting from the adoption of improved agricultural land management (ALM) practices. Eligible ALM practices include, but are not limited to reductions in tillage practices, use of organic fertilisers, residue management, water management, N-fixing crop planting and harvesting.

Inclusion Criteria for new project activity instances

The following table describes the minimum general eligibility criteria to be satisfied by new instances and evidence for how every criterion is fulfilled as per paragraph 3.1 of VCS Standard v4.2.

S. No.	Eligibility	Description
1.	Meet the applicability conditions set out in the methodology applied to the project.	As per VM0042 methodology each new project activity instance must demonstrate compliance with the applicability conditions set out in the methodology.
2.	Use the technologies or measures specified in PD	All the new project activities shall implement one of the activities mentioned in the project activity description in Section 1.11 of the PDD
3.	Apply the technologies or measures in the same manner as specified in the project description.	Each project activity instance shall adhere to the technology specifications (mentioned in methodology / standard) employed by project activity.
4.	Geographical boundaries: all new project activity instances must be located within the boundary of the grouped project.	Each project activity instance included must be within the boundaries of the grouped project.

1.5 Project Proponent

Organization name	Varaha ClimateAg Private Limited
Contact person	Ankita Garg
Title	Cofounder and CEO
Address	NO. 52, F/F, MASJAD LANE, BHOGAL, NEAR GURUDWARA, SOUTH DELHI – 110014, DELHI, INDIA
Telephone	9619014971
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1.6 Other Entities Involved in the Project

Left blank intentionally as per Section 3.1.3 of the VCS Procedural Document; Registration and Issuance Process: (Version 4.1).

1.7 Ownership

Land ownership: Majority of agricultural lands in India are owned directly by individual farmers. These farmers have well-defined land records that are registered with their state governments and can be confirmed using state government land record databases. The ownership is clear and there is no dispute over land ownership in the project site. The practices being adopted on farmer lands under the <Program Name>. The credits generated using the program belongs to the owner of the land. The ownership records are available online at <https://jamabandi.punjab.gov.in/> (Punjab), <https://jamabandi.nic.in/> (Haryana), and https://upbhulekh.gov.in/public/public_ror/Public_ROR.jsp (Uttar Pradesh). These land certificate records are also available with the Tehsildars of the respective states.

Credit ownership: During the project preparation, VarahaAg Private Ltd. (hereinafter “project proponent”) invested and cooperated with the local implementation partners for adoption of the regenerative agricultural practices. Furthermore, as the owner of the farmlands, the farmers have the right to authorize VarahaAg as a representative of the carbon project owner. Hence, VarahaAg has overall control and responsibility for the project as the project proponent without violating farmer’s right. Also, the project proponent has signed a contract with the landowner that the proponent would have the sole right to generate the credits from the referenced farmland. They would have the end-to-end responsibility to enable verification, audit, and issuance. The landowner commits to not having sold in the past, to not sell in future, any GHG reductions or removals from regenerative agriculture practices to any party other than the proponent and has agreed to provide all its legal rights under any applicable carbon standard.

1.8 Project Start Date

According to VCS standard, v4.2, the project start date is the date on which activities that lead to the generation of GHG emission reduction or removals are implemented. (Eg. Preparing land for

seedling, planting, changing agricultural practices). The start date of the project is 1st June 2019 as the implementation of several improved agricultural practices including direct seeding or rice, no-till farming, efficient nutrient, and water management have been followed for the rice crop from this date.

1.9 Project Crediting Period

The project crediting period is 20 years from the start date. The project period is thus 1st June 2019 to 31st May 2038.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☒ >1,000,000 tCO₂e/year

Project Scale	
Project	NA
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2019-2020	28122
2020-2021	50212
2021-2022	93576
2022-2023	191084
2023-2024	500439
2024-2025	1564032

2025-2026	3367788
2026-2026	5068024
2027-2028	7088425
2028-2029	7088425
2029-2030	7088425
2030-2031	7088425
2031-2032	7088425
2032-2033	7088425
2033-2034	7088425
2034-2035	7088425
2035-2036	7088425
2036-2037	7088425
2037-2038	7088425
2038-2039	7088425
Total estimated ERs	9,59,24,376
Total number of crediting years	20
Average annual ERs	47,96,218.8

1.11 Description of the Project Activity

The project is not located within a jurisdiction covered by a jurisdictional REDD+ program. The key objective of the project is to enable mitigation of greenhouse gas emissions by implementing regenerative agriculture practices. Developing farm-level adoption of climate-resilient production systems for rice, wheat, maize, cotton, mustard and sugarcane crops in the project area is critical for empowering small holder farmers in the region to sustain their livelihood and ensure income security.

Smallholder agriculture is hindered by several inter-dependent factors such as low soil fertility, frequent dry spells, drought (climate change) and unsustainable agri-management practices. Ongoing and traditional agricultural practices, such as continuous monocropping, excessive use of synthetic fertilizers beyond the recommended dosage, poor soil, crop residue and irrigation management in the project locations have deteriorated soil productivity to the extent that many

of these agricultural soils are depleted of nutrients and unable to sustain crop productivity naturally. Improving crop productivity, soil fertility via improving Soil Organic Carbon (SOC) and providing economic benefits to these smallholder farmers is a crucial challenge. The project attempts to address this challenge sustainably, by combining three founding principles of conservation agriculture, such as minimum mechanical soil disturbance, permanent soil cover, and crop rotation (FAO, 2012). Regenerative agriculture practices are reported to improve soil quality, optimize crop yields, and reduce input costs.

VarahaAg is implementing the project activities together with ground implementation teams and partners. As part of the project implementation process, field protocols are currently being developed and will be provided during verification. Further comprehensive information on field management and other aspects of the project implementation will be made available during validation processes.

Following are the indicative and most influential (in terms of Soil health/SOC, crop residue management and GHG emissions) regenerative agricultural practices adopted by the farmers based on their own crop needs and climatic conditions. Complete description of these practices and additionality assessments of any new practices will be provided during verification.

Table 1: Different Sustainable agricultural practices adopted and their benefits.

Adopted Practice	GHG Objectives	Additional Benefits
Zero tillage and dry direct seeded rice (DSR) practice	Sequestration, emission reduction	Improves soil, water, and air quality. Reduces soil erosion
Efficient nutrient management and Crop residue management	Sequestration, emission reduction	Improves water quality. Saves expenses, time, and labor.
Crop diversity through rotations and cover crops	Sequestration	Reduces erosion and water requirements. Improves soil and water quality.

- **Minimum/Zero tillage [ZT]/Reduced Tillage [RT]**

Zero tillage (ZT) implies planting crops in previously unprepared soil and is mainly considered a farm management system. It is also known as zero till, no till or direct planting. The modern concept of ZT tends to imply seeding a crop mechanically in undisturbed soil-covered plant residues (Ekboir 2002). Continuous rice-wheat-maize cropping systems cause build-up of pests and diseases. Reducing soil movement through ZT serves as an effective control measure for pests (Malik et al. 2002). Rice-wheat-maize systems also cause land degradation, with excessive groundwater pumping. Rice-wheat systems need to enhance their cost competitiveness in the context of trade liberalization. Zero tillage potentially enables savings in energy (electricity, fuel), water, labor, and other inputs such as use of machinery, costs of tillage operation, labor while greatly conserving soil moisture. The advantages of ZT technology are thus manifold generating higher yields at lower production costs and, being an environment- friendly practice that saves water and soil (Hobbs et al. 1997).

- **Direct Seeded Rice (DSR)**

Direct Seeded Rice (DSR) systems have significant potential to reduce the environmental footprint and land degradation (Yadav et al., 2020). DSR is a suitable alternative to advance climate-resilient methods in an efficient manner. The DSR option has the advantages of direct seeding in low residue pressure, using the zero-tillage and multi-crop planter, and in high residue pressure, with the Happy Seeder and Super Seeder. Furthermore, the DSR option has been known to sustain soil systems and increase productivity and profitability (Samal et al., 2017). In addition, adoption of the DSR option advances seeding by 15 days (Rao et al., 2021).

- **Crop residue management**

The project activities aim to eliminate crop residue burning, as harvesting of various crops generates large volume of residues both on and off farm. The generation of crop residues is highest in Uttar Pradesh (60 Mt), followed by Punjab (51 Mt) and Maharashtra (46 Mt). Among different crops, cereals generate maximum residues (352 Mt), followed by fibres (66 Mt), oilseeds (29 Mt), pulses (13 Mt) and sugarcane (12 Mt). Cereal crops (rice, wheat, maize, millets) contribute 70%, while rice crop alone contributes 34% to the crop residues. Sugarcane residues consisting of top, and leaves generate 12 Mt, i.e., 2% of the crop residues in India.

In the project locations, a large portion of crop residue (about 116 MT) (Porichha et. al., 2021) is recorded to be burnt 'on-farm' primarily to clean the field for sowing the next crop. The problem of 'on-farm' burning of crop residues is intensifying in recent years due to shortage of human labour, high cost of removing the crop residue from the field and mechanized harvesting of crops. As per available estimates, burning of crop residues is predominant in four states, namely, Haryana, Punjab, Uttar Pradesh & West Bengal.

In the current project, farmers who have now stopped burning are being enrolled for carbon revenues because of preventing the CO₂ and CH₄ emissions because of burning activity. Preventing burning itself would aid in SOC increases as the Carbon losses through burning will be prevented. The crop residues could also be subjected to aerobic/vermicomposting or added as mulch to improve SOC in the soils. Alternatively, the recently popularized farm machinery such as Happy Seeders and Super Seeders could be used for direct incorporation of residues.

- **Farm mechanization**

The Happy Seeder is a tractor-mounted machine that cuts and lifts rice straw, sows' wheat into the soil, and deposits the straw over the sown area as mulch. Super Seeder is another farm machinery which is considered as a single-pass solution to fulfil current farming needs to stop crop residue burning. It can be used to plough paddy stubble back into soil and sow wheat seed simultaneously in a single operation after the harvesting of the paddy. This enhances soil fertility and carbon sequestration.

Table 2: List of specific project activities being implemented in the project locations

Crop	Crop specific Regenerative practices	Common Regenerative practices
Rice	• Alternate wetting and drying (AWD) • Dry Direct seeded rice (DSR) • Wet DSR • Mechanical rice transplanter	• Water/irrigation management • No or reduced Tillage

	• System of rice intensification (SRI) • Intensification and diversification of rice cropping systems	• Relay cropping • Crop residue management (including composting)
Wheat	• Zero tillage (ZT) • Happy Seeder (HS) • Mulching • Laser land levelling (LLL)	• Organic amendments (FYM) • Cover crops (Green manuring)
Maize	• Irrigation scheduling based on soil matrix potential (SMP) • Crop diversification • Permanent Raised bed (PRB) planting	• Organic mulching • Nutrient management (fertilizer management) • Application of Nitrification inhibitors/Slow-release fertilizers
Cotton	• Cotton legume intercrop • Drip irrigation systems • Integrated nutrient management • Ratooning • Zai pits • Rainwater harvesting systems	• Crop rotation • Double cropping • Cover cropping • Intercropping (Crop diversification)
Mustard	• Mixed cropping systems • Zero/reduced tillage • Ridge transplanting • Bed Planting • Green manuring with sesbania • Biofertilizers and bio stimulants	• Farm mechanization for crop establishment • Laser land levelling • Rotavator • Laser land levelling
Sugarcane	• Surface drip and sub-surface drip irrigation practices • Reduced tillage • Urease inhibitors • Improved fertigation • Trash shredding and mulching • Composting and biofertilizer • Sugarcane-legume cropping • Crop rotation • Row-row wide spacing • Crop residue management • Green manuring	

Carbon revenues upon implementation of the above-mentioned regenerative practices will allow wide-spread adoption and recovery of soil fertility along with restoration of soil organic carbon. Soil carbon storage that results from this improved crop/soil management is expected to result in significant removal of CO₂ from the atmosphere, as will be documented using the VM0042 – Methodology for improved agricultural land management.

In order to achieve project goals, Figure 1 provides step by step approach for VarahaAg's Carbon farming program. While scaling up of regenerative agriculture practices is in progress in the project locations, VarahaAg is concurrently in the process of developing digital platforms for ground truthing as well as farmer enrolments (providing detailed information about their farming activity data through applications) which will be made accessible on smartphones. Once the ground data is received, VarahaAg will process the data provided by farmers and will be

modeled/analyzed further to quantify GHG emissions and the SOC sequestered in the project activities registered and approved for the MRV plans.



Figure 1: Representation of carbon farming program being implemented by the Project Proponent.

1.12 Project Location

Project is currently being implemented in the region of Indo-Gangetic plain (IGP). States included in the region are Punjab, Haryana, Uttar Pradesh, Uttarakhand, Bihar, West Bengal and Orissa. The below map shows the project state where project is implemented.

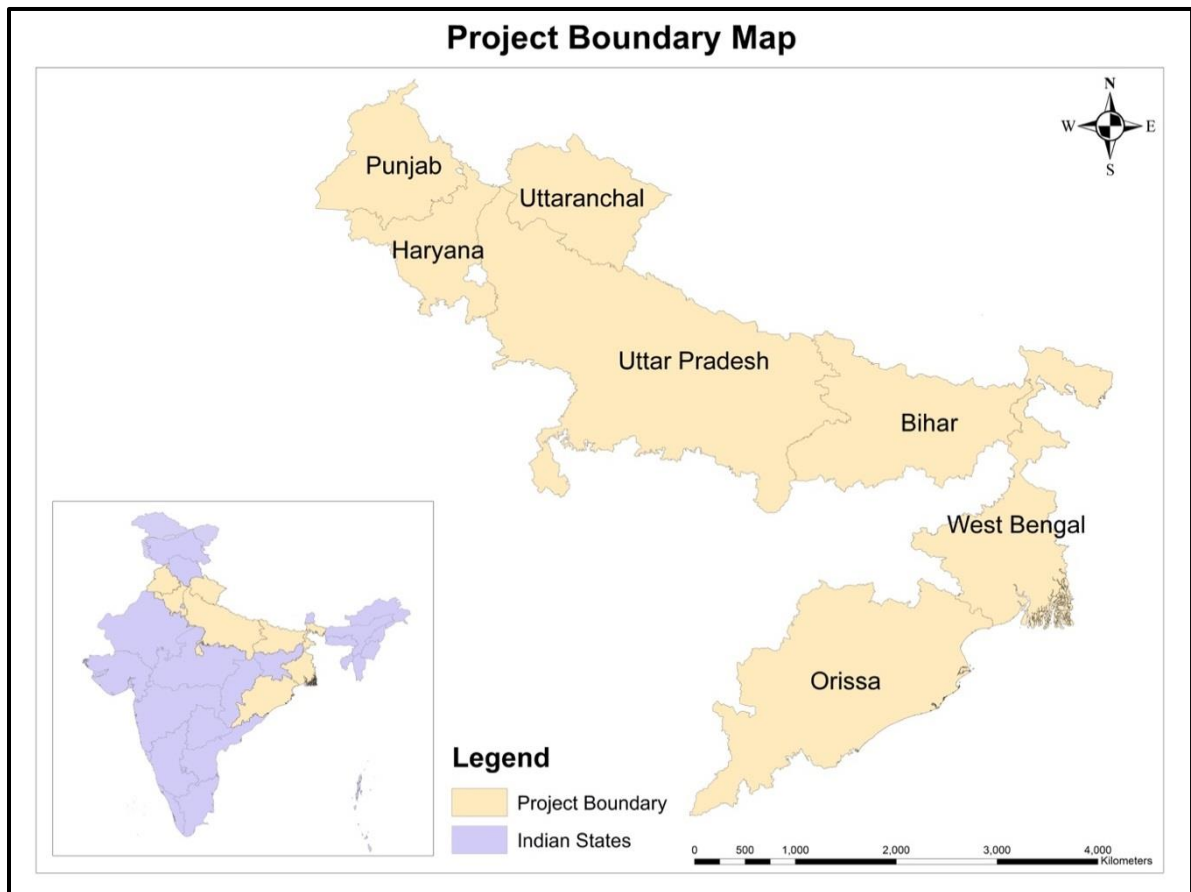


Figure 2: Map showing the states in which project will be implemented.

1.13 Conditions Prior to Project Initiation

Ecosystem type: The ecosystem type prior to the project initiation was basically terrestrial ecosystem with land classified as agricultural lands.

Current and historical land-use: The historical land-use of the lands selected under the project activity was agricultural activities. Land Use Land Cover (LULC) map for the project location has been prepared (Figure 2) to prove the project location is the crop land or agriculture land.

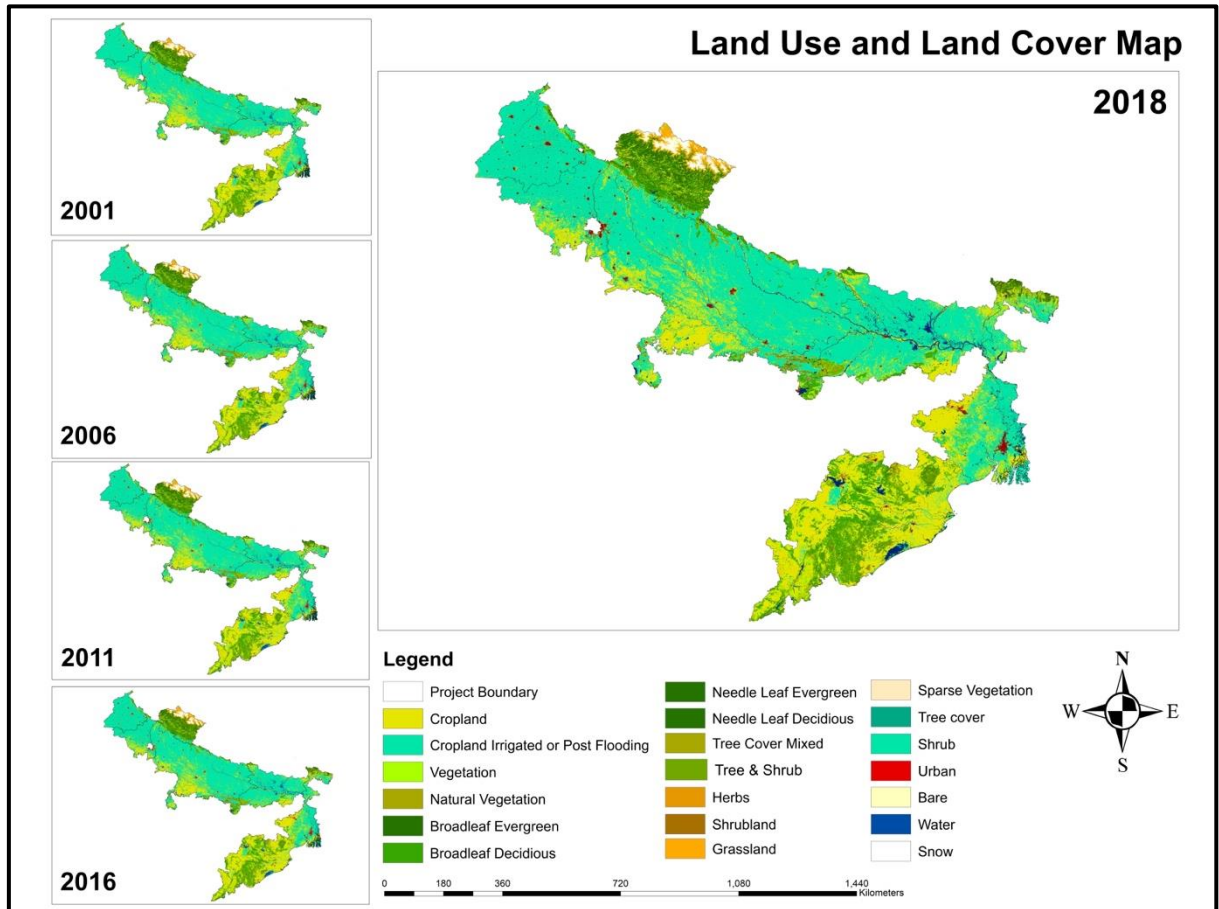


Figure 3: Historic LULC map of the Project Location (Indo-Gangetic Plains).

Climatic conditions: The IGP is found in subtropical to moderate temperate climates, with chilly and dry winters and hot and humid summers (Timsina and Connor 2001). The upper-IGP receives roughly 550 mm of yearly rainfall (low), whereas the lower-IGP receives 1200 mm (high).

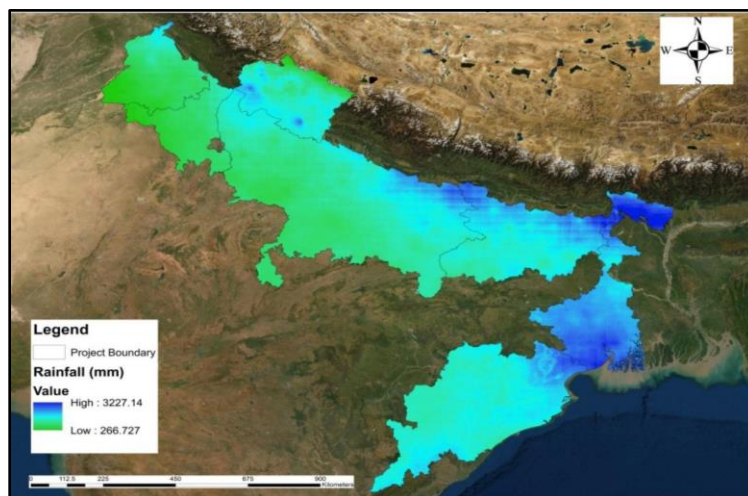


Figure 4: Annual rainfall Map of the Project Location (Indo-Gangetic Plains).

Hydrology: The Indo-Gangetic Plain (IGP) is divided into two drainage basins: the Punjab and Haryana plains in the west, and the Ganges–Brahmaputra drainage systems in the east. Water from the rivers Ravi, Beas, and Sutlej, Yamuna and Ganga is used to irrigate the plains. With key crops such as rice and wheat, the Indo-Gangetic Plains are the world's most intensively farmed territory. Other crops farmed in this region include maize, sugarcane, and cotton.

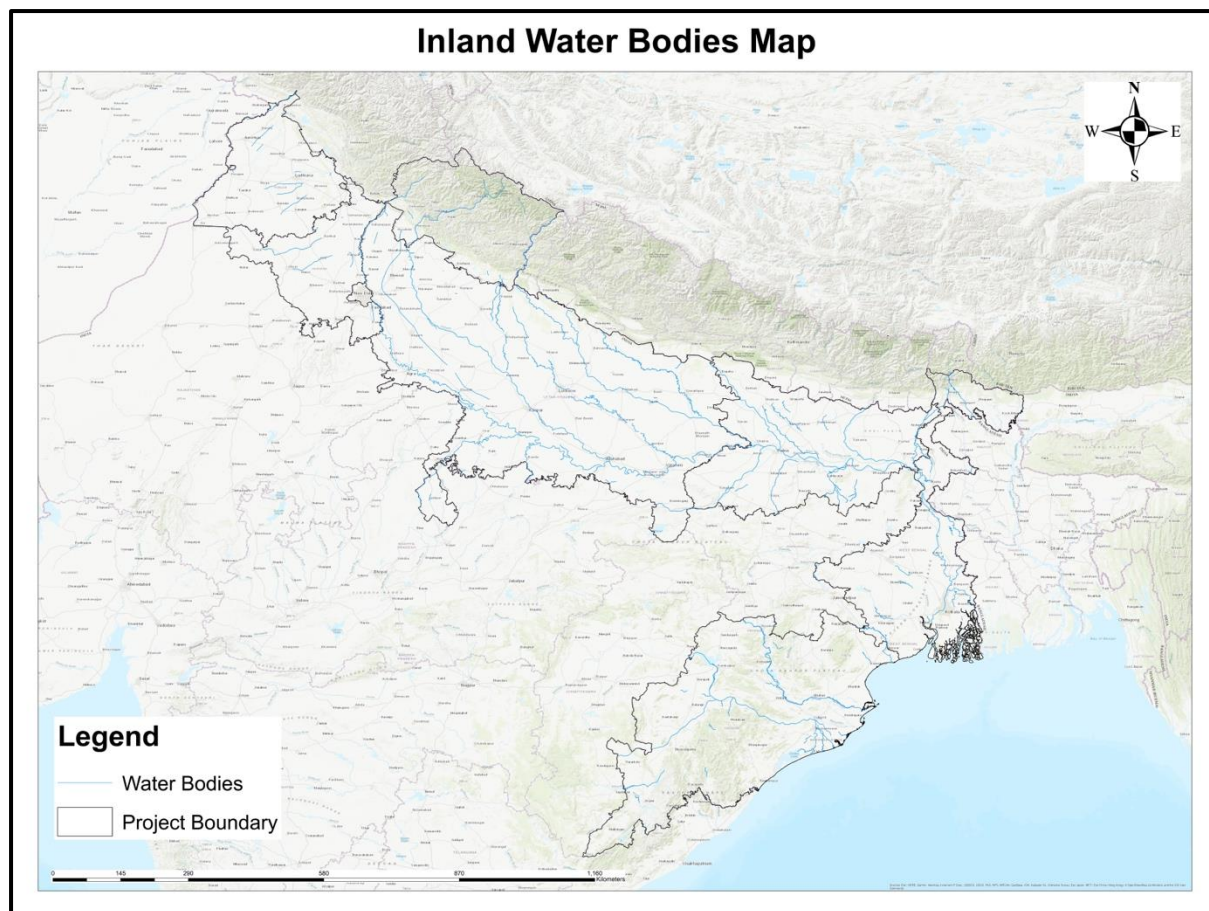


Figure 5: Inland water bodies Map of the Project Location (Indo-Gangetic Plains)

The Indo Gangetic Plain (IGP) is one of the world's most densely populated locations, owing to its agriculturally productive soil. In terms of vegetation, the IGP is a reasonably similar biological zone, however it has been separated into five broad transects based on physiography and bioclimate (Narang and Virmani 2001). The Trans-Gangetic Plains (transects 1 and 2) cover huge portions of Pakistan and India, including Punjab and Haryana. The areas in Uttar Pradesh, Bihar, and Nepal are covered by Transects 3 and 4. Transect 5 includes the lower Gangetic Plains in West Bengal (India) and parts of Bangladesh.

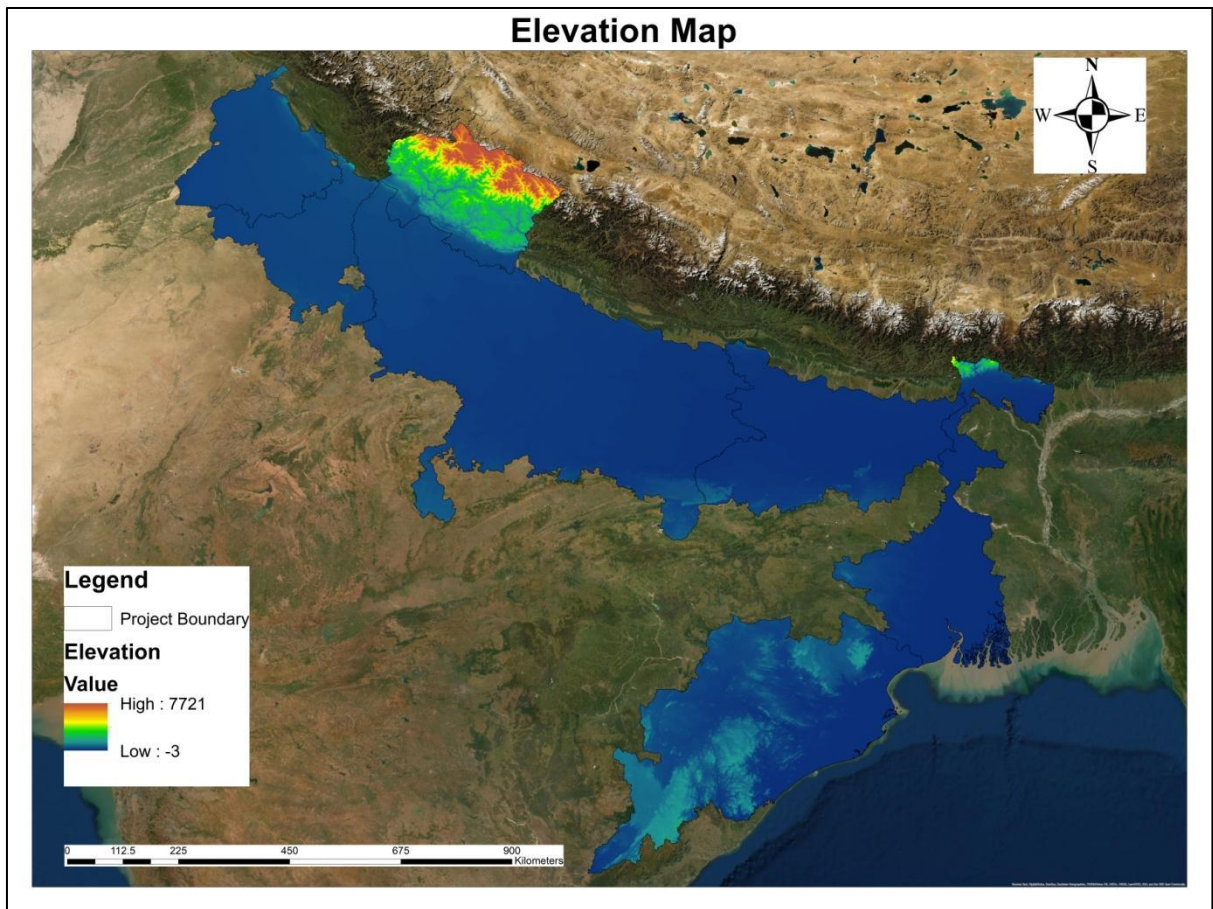


Figure 6: Elevation map of the Project Location (Indo-Gangetic Plains)

Soil Health: Intensive use of irrigation water in rice led to increased salinity. In the short term, salinity built up leads to reduced yields, whereas, in the long term, it can lead to abandoning of crop lands. Also, the heat from burning paddy straw penetrates 1 cm into the soil, elevating the temperature to 33.8 to 42.2 degree Celsius. This kills the bacterial and fungal populations critical for a fertile soil. Burning also causes damage to other micro-organisms as well as its organic quality. One tonne of stubble burning leads to a loss of 5.5 kg N, 2.3 kg P, 25 kg K and more than 1kg of Sulphur. Therefore, conventional agricultural practices and crop burning practices are proving to be detrimental to soil health, crop productivity and farmer incomes in the IGP region.

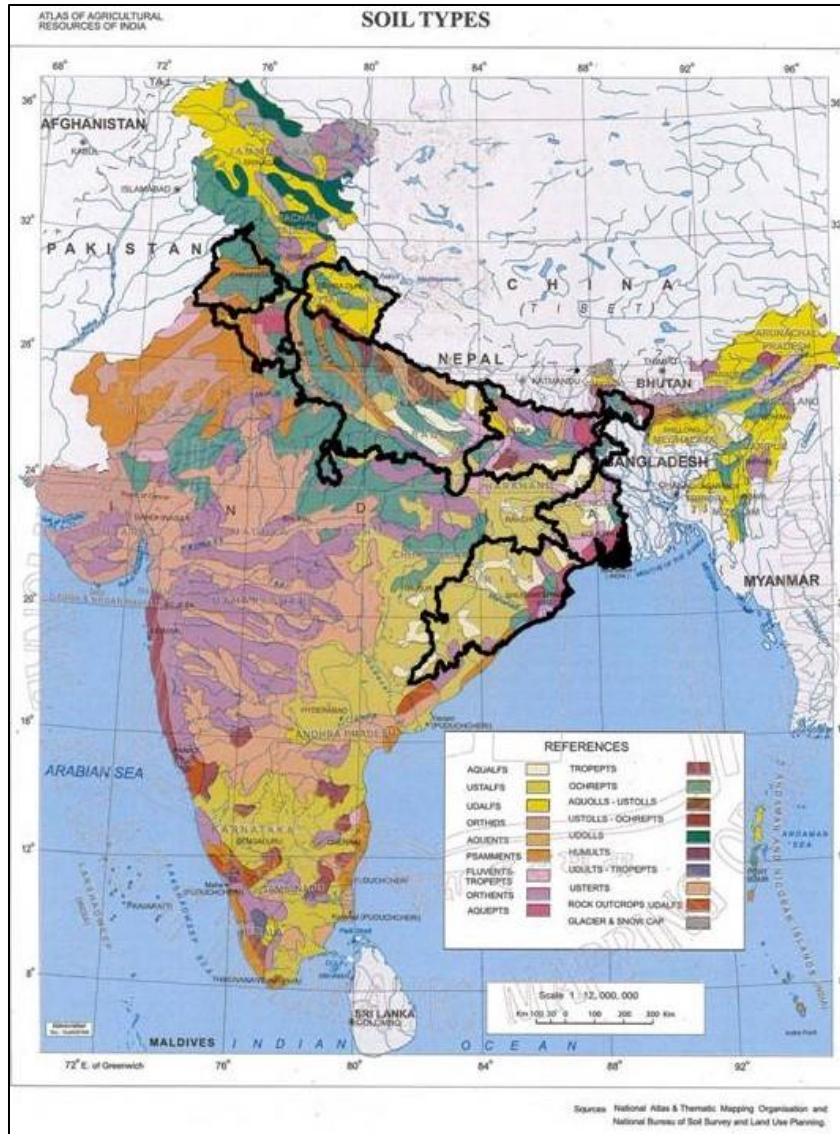


Figure 7: Soil Map of India showing the boundary of project area.

Has the land been cleared of native ecosystems within 10 years of the project start date?

☒ Yes

☐ No

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

India does not regulate GHG emissions from Agriculture. However, the project activities comply with all applicable laws, policies, and regulatory frameworks. There may be government initiatives supporting certain subsets of project activities, but there are no enforceable rules mandating the adoption or avoidance of any project activity.

However, national laws are given below, which the projects activities are not violating:

Environment Protection Act, 1986⁸ - The project complies with all rules and regulations of the Environment Protection Act. The project activities do not cause any harm, in fact improves the environment by implementing improved agricultural practices reducing the emissions.

Wildlife Protection Act, 1972⁹ - Any activity that may have an influence on wildlife is not included in the project. Hence it is not relevant, yet the project activity follows all regulations and principles of the act.

Forest Conservation Act, 1980¹⁰ - The project activity is being implemented on croplands/ agricultural lands. There is no land use change from forest land to cropland. Therefore, the law is not applicable to the project activity, but the project activity complies with all the rules and guidelines of the act.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not registered in any other GHG program other than VCS.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been submitted to any other GHG Programs and has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The project proponent itself does not have any binding limits on GHG emissions and does not look for any compliance with an emission trading program.

1.16.2 Other Forms of Environmental Credit

There is no other form of environmental credits included in the project framework or currently ongoing in the project area. The project has not participated in any program for environmental

⁸ <https://web.archive.org/web/20020613044316/http://envfor.nic.in/legis/env/env1.html>

⁹ https://legislative.gov.in/sites/default/files/A1972-53_0.pdf

¹⁰ https://legislative.gov.in/sites/default/files/A1980-69_0.pdf

crediting for GHG emission removals. The project does not intend to generate any other kind of environmental crediting for GHG emission removals

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

1.17 Sustainable Development Contributions

The United Nations adopted 17 interlinked global goals, Sustainable Development Goals (SDGs) in 2015 which are “Blueprint to achieve a better and more sustainable future for all.” There are 17 SDGs in total with each goals having their own set of targets. Niti Aayog – The Planning Commission of India, has created the SDG India Index which covers 13 out of 17 SDGs (excluding the SDGs 12, 13, 14 and 17)¹¹. The SDG India Index emphasizes on holistic approach to assess environmental, social and economic status of the country.

Project’s contribution for achieving nationally states sustainable development priorities.

SDG India Index Goal		Relevance
No Poverty		The project activities facilitate additional incomes to the small and marginalised farmers in terms of Carbon credits as an incentive for adopting sustainable and regenerative agricultural practices. Environmentally friendly sustainable agricultural practices have long term positive effects on the crop productivity and ultimately food and income security.
Zero Hunger		Improved regenerative agriculture practices enable improved crop productivity and improved income benefits, thus allowing people to afford (to buy sufficient) food
Good Health		Carbon credits generated through the project are envisioned to encourage more and more farmers for adoption of prescribed crop residue management practices and contribute to complete elimination or stopping of burning practices. This will lead to prevention of pollution due to toxic gaseous emissions (CO ₂ , CH ₄ and CO, etc.) into the environment and thus contribute to better health for human and animal life.

¹¹ <https://pib.gov.in/PressReleasePage.aspx?PRID=1723952>

SDG India Index Goal		Relevance
Responsible Consumption		The project enables sustainable management and efficient use of natural resources. The project also promotes environmentally sound management of chemical fertilizers applied to the soil, in accordance with national guidelines, and significantly reduces their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.
Climate action		The project enables adoption of climate smart agriculture practices by the farmers preventing the climate change related adversities to their crop production and building resilience to any unforeseen climatic situations. The project activity also enhances carbon sequestration through integration of cover crops etc.
Life on Land		The project ensures the conservation, restoration and sustainable use of natural resources such as water and soil and restores degraded land and soil ecosystems which have also been affected by drought/floods.

1.18 Additional Information Relevant to the Project

Leakage Management

Nil

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable.

2 SAFEGUARDS

2.1 No Net Harm

This project is not expected to result in negative environmental impacts. Instead, project activities are expected to lead to positive environmental impacts beyond GHG emission reductions and removals which has been confirmed during the stakeholder consultation meeting with the farmers.

Negative socio-economic impacts at the community level are not anticipated. At the farm level, farmers may experience financial setbacks in initial years due to upfront costs of practice adoption and changes to yield as farmers identify the best ways to implement new practices on their lands. Any potential economic impacts are expected to be minimal and temporary.

Additionally, financial support to farmers through payments and sale of carbon credits are intended to counterbalance any initial increased expenses or changes to revenue. Over the years Project Proponent expects farmers to see financial gains from increased yields in regular and, especially, extreme weather years due to increased soil health and overall farm resilience.

2.2 Local Stakeholder Consultation

The local stakeholder meetings were organised by Varaha Climate Ag Pvt. Ltd. with smallholder farmers, and project partners (FPOs, Rice producing companies) in different areas of Haryana and Punjab to apprise smallholder farmers about the carbon project development. The main aim of the stakeholder meeting was to inform farmers, project partners, FPOs (Farmer Producer Organisations) about the project and its design and give them an opportunity to discuss and give feedback on the project and its potential environmental and social impacts. The overall goal was to improve the project design based on stakeholders' comments and suggestions and increase the local awareness and involvement in the project.

Stakeholders consultations were held at Karnal in Haryana, Gurdaspur and Ropar in Punjab on 28th July, 2022 and 8th -10th August, 2022 respectively. Furthermore, the Project Proponent is currently conducting similar stakeholder consultations in various districts of the project area. This will be covered in detail in the Final Project Description.

The meeting was attended by the following participants-

1. Dr. OP Lathwal, Regional Director, Rice Research Centre, Kaul, Haryana
2. Dr. Kanwalpreet Singh - Chief Agriculture Officer (Gurdaspur)
3. Dr. Sandeep Singh - Project Director (ATMA)
4. Dr. Prabhjot Singh (Deputy project director- ATMA)
5. Dr. Sumeesh Arora (KVK Farm Advisor)
6. Ms. Ankita Garg, Chief Operating Officer (COO), Varahaag
7. Ms. Supriya Kumari, Manager - Carbon Projects, Varahaag
8. Mr. Vikarmaditya Yadav, Manager - Carbon Credit Supply, Varahaag
9. Ms. Chanchal, Manager - Carbon Projects, Varahaag
10. 6 members from the field team of Varahaag
11. 10 representatives from different FPOs
12. 54 farmers from Haryana
13. 60 farmers from Punjab



Figure 8: Banner at Haryana and Punjab for Stakeholder consultation

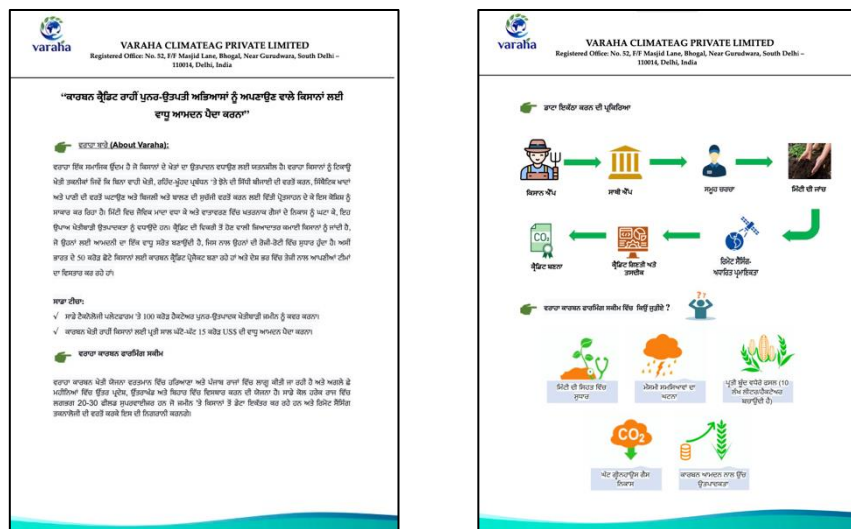


Figure 9: Manual in local language had been prepared for local stakeholders imparting the details of carbon project in their area

Welcome Address by Ms. Supriya, Manager Carbon Projects, Varahaag

Ms. Supriya welcomed the chief guest Dr. OP Lathwal, the FPO representatives, and the farmers in the stakeholder meeting, giving a brief about the purpose of the meeting as well as about the agenda of the stakeholder. She introduced Varahaag to the participants, as a social enterprise working on combating climate change through regenerative agricultural practices and helping smallholder farmers to improve their livelihood by providing additional income. She also highlighted the benefits of carbon farming to the farmers, focusing on the fact that farmers need not to invest a single penny at any stage of this carbon finance project development.



Keynote address by Dr. OP Lathwal, Rice Station Head, Kaul, Karnal

Welcoming the Varahaag team and the participants, Dr. Lathwal talked about the connection of this program with the farmers. He highlighted the importance and expansion of agriculture in all the different sectors and how different organisations are working to improve the livelihood of small farmers. Different sectors want to build a participatory approach with farmers to protect the environment, air and water. In comparison to the mid 90s, the yield as well as the soil quality has degraded. The yield has decreased from 90-95 quintal/ acre in the 1990s to 40-45 quintal/ acre and the organic carbon has decreased from 1.2 percent in the 1940s to 0.2 percent in 2022. Dr. Lathwal threw light on the major issues leading to this degradation i.e. the excessive use of fertilisers/ pesticides and the decreased microorganisms in the soil due to intensive farming practices. The physical, chemical and biological properties of the soil are damaged resulting in decrease in yield. He focussed on the adoption of conservation agriculture practices for restoring crop yield as well as improving soil conditions. He also talked about the three main principles of regenerative agriculture practices, i.e. minimum use of machinery on the farms (No-till), residue should be left undisturbed after harvesting, and crop rotation. He encouraged farmers to adopt regenerative practices and to enrol themselves in the Varaha Carbon Kheti Yojana to get additional benefits in terms of carbon credits.



Video made by Varahaag team

A small “At a glance” video was prepared by the Varaha team to provide insights of the program and to make the concept of carbon kheti easy to understand for the farmers. After the keynote address of Dr. OP Lathwal, the video was played to give them a brief about the “**Varahaag’s Carbon Kheti Yojana**”.



Presentation by Ms. Ankita Garg, Chief Operating Officer (COO), Varahaag

A detailed presentation was made by Ms Ankita Garg on “Generating additional income via carbon credits for farmers adopting regenerative agricultural practices”. She explained the negative impacts of GHG emissions and the agriculture sector accounting for 25% of total emissions. Moving on to comparison between conventional and regenerative farming practices, she explained the importance of conservative farming practices and how these practices generate carbon credits. Encouraging farmers to be a part of this program, she also talked about the data collection process and the benefits of regenerative farming for the farmers.



Presentation on Project Activities by Mr. Vikarmaditya Yadav, Manager, Carbon Credit Supply

Mr. Vikarmaditya Yadav, gave detailed information on the activities the farmers are currently doing and credits generated from each regenerative practice. For instance, Direct Seeding Rice, zero till farming and incorporation of crop residue generates 1 credit/ acre / activity. He also informed the participants about the project area, i.e. states in which the program is being implemented i.e. Haryana and Punjab and future expansion of the program. He highlighted that till now Varahaag has boarded approximately 40,000 acres of land in this project.



Feedback from the participants:

- The participants were very optimistic about the project and extended their complete support.
- Farmers suggested that such stakeholder meetings should be conducted regularly.
- Farmers agreed to assist the Varaha team during field visits and provide appropriate data required for the project.
- Farmers appreciated the environmental benefits, additional monetary benefits and productivity enhancement due to the implementation of project activities.



2.3 Environmental Impact

This project activity involves regenerative agricultural practices and comes under AFOLU (Agriculture, Forestry and Other land Use) criteria thus this project activity does not have any negative environmental impact. The project activity is completely environmentally friendly and has positive impacts on the environment. This has been confirmed during the local stakeholder consultations. The project will achieve various environmental benefits such as biodiversity conservation, increased soil fertility, water conservation, increased food production, climate change mitigation and adaption etc.

2.4 Public Comments

Public comment to be addressed in the Final PD.

2.5 AFOLU-Specific Safeguards

All the local stakeholders have been included in the consultation during the development of carbon project. Any foreseeable risk identified during stakeholders' consultation will be addressed and ensured that it would not impact the project activity. The project proponent and project instance implementer will have a grievance redressal mechanism to address any issue and resolve it sooner.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The VCS methodology “VM0042 Methodology for Improved Agricultural Land Management, Version 1.0” is being used to quantify the emission reductions¹².

Additional tools applied-

- VCS Standard, version 4.1 – 22 April 2021
- VCS Program Definitions, v. 4.1 – 15 April 2021
- VCS Program Guide, v. 4.0 – 19 September 2019
- Registration & Issuance Process, v4.0 – 19 September 2019
- AFOLU Non-Permanence Risk Tool, v4.0 – 19 September 2019
- VMD0053 Model Calibration, Validation, and Uncertainty Guidance for the Methodology For Improved Agricultural Land Management¹³
- Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities
- Simplified baseline and monitoring methodology for small scale CDM afforestation and reforestation project activities implemented on lands other than wetlands
- Tool for testing significance of GHG emissions in A/R CDM project activities

Full list of additional tools to be confirmed in final version of PD.

3.2 Applicability of Methodology

All new activities must meet applicability conditions listed in the table below and will be demonstrated with justification in the project document.

With the help of VM0042 methodology, quantification of CO₂, CH₄ and N₂O emission fluxes can be recorded, modeled and analyzed. The methodology is global in scope and applies to a broad range of agricultural management project activities that increase SOC storage and/or contribute

¹² https://verra.org/wp-content/uploads/2020/10/VM0042_Methodology-for-Improved-Agricultural-Land-Management_v1.0.pdf

¹³ https://verra.org/wp-content/uploads/2020/10/VMD0053_Model-Calibration-Validation-and-Uncertainty-Guidance-for-the-Methodology-for-Improved-Agricultural-Land-Management.pdf

to the decrease of net emissions compared to the baseline scenario. The methodology is applicable for this project due to the following reasons:

Applicability Conditions	Justification
1. Project must introduce or implement one or more new changes to pre-existing agricultural management practices; Reduced (organic or inorganic) application, improved water management, reduce tillage, improved crop planting and harvesting and improve grazing practices	Condition met. The project introduces improved regenerative agricultural practices that can replenish Soil Organic Carbon stock levels in the project locations whilst restoring and enhancing soil health/soil fertility. The details are mentioned in section 1.11.
2. The project activities must be implemented on either cropland or grassland at the project start date and remains cropland or grassland throughout the project crediting period.	Condition met. The project involves only croplands as per methodological guidelines. Our Geographic Information System (GIS) analysis has verified that the croplands remained so over the past 10 years and will remain as croplands during the project lifetime. (Section 1.13, Figure 3).
3. The project area must not have been cleared of Native ecosystems within the last ten years prior to the project start date:	Condition met. The project will focus on lands that have been in the cropping system proposed for at least 10 years. (Section 1.13, Figure 3).
4. The project activity proposed is not expected to result in a sustained reduction of greater than 5% in productivity:	There is sufficient literature published in peer-reviewed journals proving that these practices positively impact the productivity. The activities under regenerative agriculture will help farmers to maintain and enhance their crop, soil, and water productivities in a sustained manner. However, crop yields will be relative to each farmer's baseline yields and the specific cropping system employed (also based on geographic location). Generally, regenerative agriculture practices can sustain as well as improve yields and/or lead to reduced crop production costs. Some practices, such as adding organic amendments to crops instead of synthetic fertilizer, have been reported to directly increase crop yields. For other practices, it has been observed that yields might be impacted to $\leq 10\%$ in the first 2-3 years of adoption and will improve over time.
5. If the project activity involves the application of biochar, it must be produced using feedstock that would otherwise have been left to decay in aerobic or anaerobic conditions or been burned in an uncontrolled manner:	If used, only the biochar produced using feedstock that would otherwise have been left to decay in aerobic or anaerobic conditions or burned uncontrolled will be considered in the project activity.

The methodology also uses the following CDM tools-

Tools	Justification
Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities	The project uses this tool for estimation of carbon stocks and change in carbon stocks of trees and shrubs in the baseline and project scenarios.
Tool for testing significance of GHG emissions in A/R CDM project activities	The project applies this tool to determine GHG emissions sources, possible decreases in carbon pools, and leakage emissions are insignificant.

3.3 Project Boundary

The Grouped Project and each new project instance will define a project boundary as per the VM0042 v1.0 methodology. The carbon pool included in Instance 01 is outlined in the table below:

S. No.	Carbon pool	Included?	Justification
1.	Above ground woody biomass	No	Project activities does not involve agroforestry
2.	Above ground non- woody biomass	No	As per VCS rules this has been excluded. (VM0042, table 2)
3.	Below ground woody biomass	No	Project activities does not involve agroforestry
4.	Below ground non- woody biomass	No	As per VCS rules this has been excluded. (VM0042, table 2)
5.	Dead wood	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature, per the VCS rules.
6.	Litter	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature, per the VCS rules.
7.	Soil Organic Carbon	Yes	Included as per VCS AFOLU Requirements as the land cover under the baseline scenario is comprised of croplands. The proposed project activity increases this pool in the project scenario.
8.	Wood products	No	As per VCS rules this has been excluded. (VM0042, table 2)

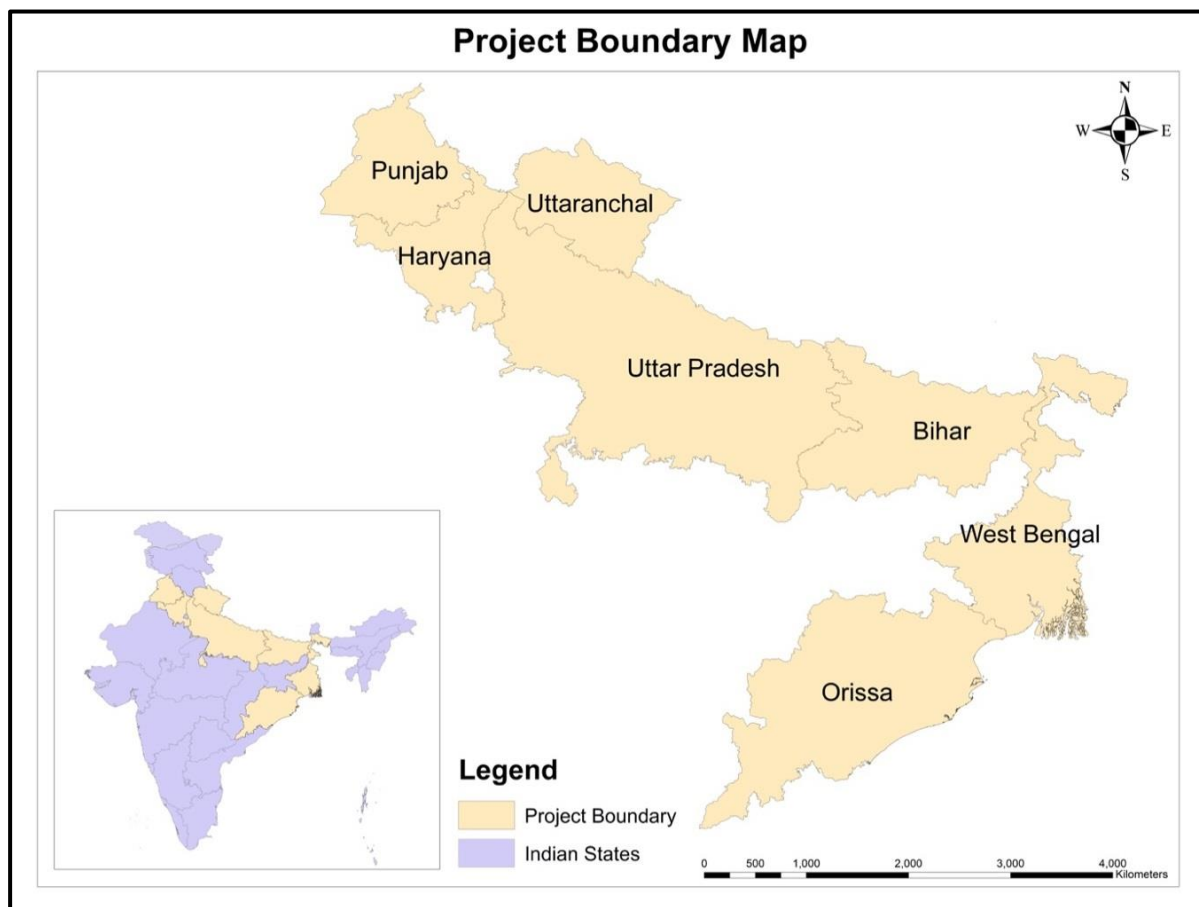
The project area includes all the farms where sustainable agricultural practices have been adopted. The project area includes cropland in the Indo gangetic plains. The details have been provided in section 1.12.

The relevant GHG sources, sinks and reservoirs for the project and baseline scenario are given in the table below:

Sources	Gas	Included?	Justification/Explanation
Soil Organic Carbon	CO ₂	Yes	Soil Organic carbon is a direct source of CO ₂ into the atmosphere. Our project will record the changes in the SOC pool. The activities taken up in the project will have an impact on the SOC pool.
Fossil Fuel	CO ₂	Yes	The regenerative practices taken up as a part of the project reduce the amount of fossil fuel. These fuels are used during various process of farming

Sources	Gas	Included?	Justification/Explanation
			such as (Land Preparation, tillage, irrigation, post-harvest management etc.)
Soil Methanogenesis	CH ₄	Yes	Certain activities such as Direct Seeding of Rice lead to a reduction in soil methanogenesis and thus the CH ₄ released in the process as compared to the traditional practice of transplanting.
Enteric Fermentation	CH ₄	Conditional	If the livestock fall under the project boundary and is grazed on the field then Enteric Fermentation will be considered
Manure Deposition	CH ₄ and N ₂ O	Conditional	If the livestock fall under the project boundary and is grazed on the field then manure deposition will be considered
Use of Nitrogen Fertilizer	N ₂ O	Yes	The regenerative practices taken up as a part of the project reduce the amount of synthetic nitrogen fertilizer and promote the use of organic manures. Thus, the usage of fertilizers will be taken into consideration for Baseline and project boundary.
Use of Nitrogen Fixing species	N ₂ O	Yes	The project activities promote the usage of nitrogen fixing species as cover crops and intercrops. these decrease the release of nitrous oxide into the environment.
Biomass Burning	CH ₄ and N ₂ O	Optional	Burning will be taken as per the data received.
Woody Biomass	CO ₂	Excluded	Woody Biomass is excluded since the activities taken up in the project don't have a significant impact on woody biomass

The geo-reference boundaries of the project areas will be clearly delineated and will be available at any time when new areas are added.



3.4 Baseline Scenario

As per VCS VM0042 methodology, the most plausible baseline scenario is that each farmer continues to manage their field with the same conventional practices they were using before the implementation of the project activities. It is distinct for each farmer and sample unit. Typically, the baseline scenario for project instances includes conventional tillage practices that disturb soil, leaving minimal residue on the soil surface.

The baseline scenario is formally defined by developing a schedule of activities existing prior to the project implementation for a period of 3 years or longer. Using historical management information from a sampling unit, the project proponent will establish the baseline schedule of activities. This will be the basis of modelling changes in soil organic carbon stocks in the dynamic baseline ex post following VM0042's Quantification Approach 1. Historical management activities also define fertilizer, fossil fuel combustion, and cropping activities resulting in N_2O , CH_4 , and/or CO_2 emissions, which are calculated for the baseline scenario using emission factors and rates or values derived from historical management information under Quantification Approach 3.

The schedule of activities described in Table 4 of VCS VM0042 will be collected for each participating farmer. Minimum specifications on agricultural management practices (for example) for the baseline scenario are-

1. Crop planting and harvesting- Major crop include Rice, Wheat, Millets, Pulses, Beans
2. Nitrogen fertilizer application- All the farmers apply synthetic nitrogen fertilizers, such as urea prior to the start of cultivation practices.
3. Tillage- Tillage is always used for land preparation on agricultural lands.
4. Residue management- As a part of land preparation, farmers burn residue.
5. Irrigation- Farmers use irrigation facilities during the dry season. Irrigation is rainfed during the monsoon season.
6. Grazing Practices- Grazing activities are not present because the project includes cropland only. Hence, the activity is not present in business-as-usual condition.
As a result, in the context of our project, it is insignificant.

The project proponent will follow VCS VM0042 requirements for how to gather evidence of qualitative versus quantitative aspects of agricultural management activities, using secondary data, farmer attestation, and remote sensing tools. Qualitative data will be determined via consultation group of expert farmers, focus group discussion (FDGs), farmer survey questionnaires etc. The collected information further will be validated by emerging technologies like Remote sensing and discussions with the group of expert farmers/agronomist. Since, VM0042 permits use of emerging technologies like remote sensing, they can be used to reliably determine quantitative information.

The figure below illustrates the existing challenges in the project area and corresponding solutions using carbon finance project. The regenerative agricultural practices adopted in the Project scenario will lead to the generation of carbon credits, which will be an additional income for the farmers for continuous adoption of these practices.

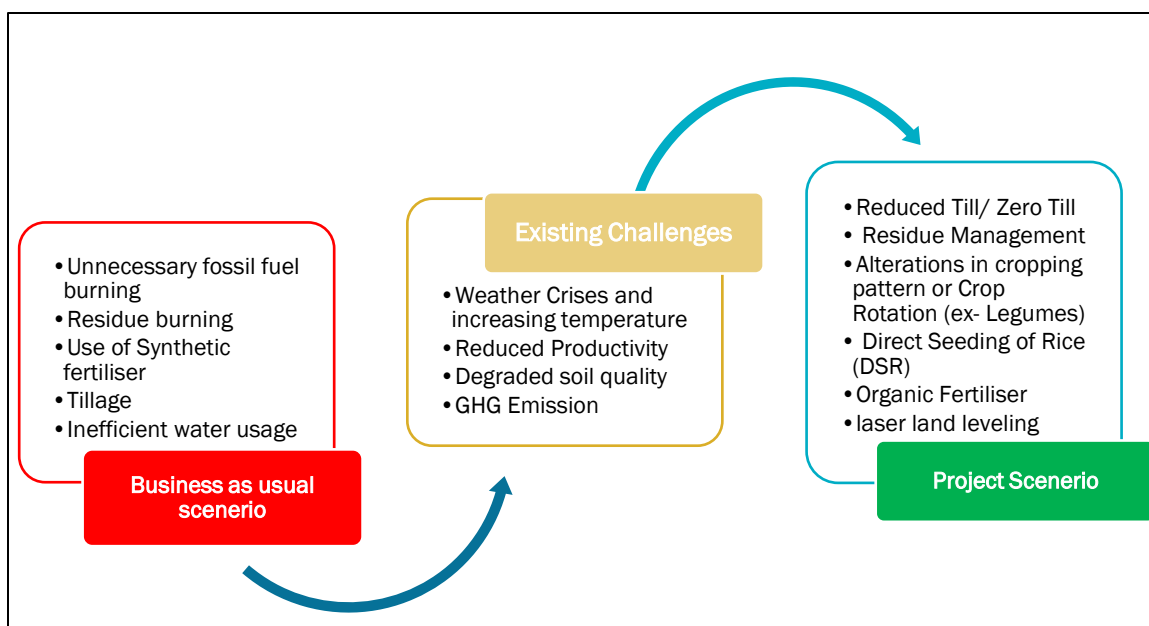


Figure 10: Adapted from CCAFS South Asia and Aggarwal et al., (2013)¹⁴

3.5 Additionality

Additionality is demonstrated in accordance with the VCS methodology requirements v4.1, section 3.5.3, and using Project method as specified in methodology VM0042.

Regulatory Surplus

At present, the Project proponent has not identified any regulatory framework, enforced law or any other regulations in the project area, which governs the regenerative agricultural practices adopted by the participating farmers. Thus, the project fulfils the regulatory surplus requirements of the methodology. In addition, the project proponent will assess and record any existing law, or regulatory frameworks that mandates the project activity in new project instances as well.

Other than the regulatory surplus, additionality evaluation will be carried out in the following two steps prescribed by the VM0042 methodology:

1) Identify barriers that would prevent the implementation of a change in pre-existing agricultural management practices

The implementation of regenerative agricultural methods has substantial technical, financial, knowledge, and cultural barriers, as most of the farmers have limited access to both technical and financial resources as well as information necessary for implementation of these methods.

¹⁴ Aggarwal, P. K., Zougmore, R. B., & Kinyangi, J. (2013). Climate-smart villages: A community approach to sustainable agricultural development.

Technological Barriers: Farmers are practising common conventional/ traditional practices since very long. They have limited technical knowledge related to the reduced fertiliser intake, reduction in tillage, improvement in crop residue, soil testing and soil nutrient management etc. Therefore, farmers were not able to change the traditional practices. Project Proponent through the project partners will impart training to the farmers regarding the regenerative agricultural practices to make them aware and will provide continuous technical support during the project lifetime. Hence, without the interventions of the project proponent the practices would not have been changed.

Financial Barriers: There are a number of schemes and subsidies in India, for farmers adopting different practices, or for the equipment's used. But these funds are either insufficient¹⁵ or the farmer is unable to get the money on time in order to execute the practices in sustainable and timely manner. Moreover, farmers are unable to buy the expensive machineries and equipment's and hence they have not changed their traditional practices. Revenue generated from the sale of carbon credits along with overall increment in crop in productivity will overcome the financial barriers due to the higher costs and a lack of suitable incentives.

Cultural and Social Barriers: There are a number of beliefs associated with different practices, like burning of crop residue helps control pests and diseases, tillage improves soil productivity etc. In addition, a large number of farmers have poor literacy rate having limited access to networks connecting them with higher level of governance and agricultural experts. Due to the implementation of the project, technicians and experts will visit the farms regularly to help farmers in order to adapt the regenerative agricultural practices.

2) Demonstrate that the adoption of the suite of proposed project activities is not common practice

For *grouped* projects with *diverse* project activities, VM0042 requires *that* project proponents demonstrate that the weighted average adoption rate of these *project* activities is less than 20%. This weighted average adoption rate is calculated based on the project area implementing each regenerative practice or combination of *practices* using publicly available data. It has been reported that less than 4 per cent of Indian farmers have adopted sustainable agricultural practices and systems, according to a study done by the Council on Energy, Environment and Water (CEEW)^{16,17}. The study, supported by the Food and Land Use Coalition (FOLU), found that scaling up sustainable agriculture would be critical to improve farm incomes and bolster India's nutrition security in a climate-constrained future. Most conservation agriculture practices are being adopted by less than five million (or four per cent) of all Indian farmers and many are practised by less than one per cent (CEEW)¹⁸.

¹⁵ [Ending burning of crop stubble through Happy Seeder technology - Tata Trusts](#)

¹⁶ https://economictimes.indiatimes.com/news/economy/agriculture/less-than-four-percent-indian-farmers-adopted-sustainable-agricultural-practices-saysstudy/articleshow/82165017.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

¹⁷ <https://www.ceew.in/press-releases/less-4-indian-farmers-have-adopted-sustainable-agricultural-practices-ceew>

¹⁸ https://www.ceew.in/sites/default/files/CEEW_ZBNF_Issue_Brief_2nd_Edition_19Sep18.pdf

Table 3: Adoption rate by area of regenerative practices

Common Regenerative practices	Area Under the Regenerative Practice (millions of hectares)	Scale of Adoption (millions of farmers)	Extent (No. of states) of adoption	Adoption Rate (By Area)
Cover crops	1.9	1.5	All	1.4%
Organic farm	2.8			
Conservation Agriculture	2	1	4-5	1.4%
Crop rotation	30	15	All the states	21%
Intercropping	1	0.8	All the states	0.7
Natural Farming	0.7	0.8	3-4	0.5%

Because of low adoption of regenerative practices (<4%) in the project location, the proposed project activities are not common practices. Therefore, the project will contribute to additionality.

3.6 Methodology Deviations

There are no deviations from the methodology adopted.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

4.2 Project Emissions

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

4.3 Leakage

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

4.4 Net GHG Emission Reductions and Removals

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

5 MONITORING

5.1 Data and Parameters Available at Validation

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

5.2 Data and Parameters Monitored

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

5.3 Monitoring Plan

This part has been omitted under section 3.1.3 of the Verra Registration and Issuance Process v4.0.

APPENDIX